

Researchers studying single-cell gene expression commonly use various clustering algorithms to group cells based on their transcriptomic similarity. When analyzing spatial transcriptomics (ST) data, this type of analysis gains an additional purpose: Identifying tissue niches or domains. Multiple options are available for clustering ST data, with some treating spots or cells as spatially independent while others explicitly incorporate spatial information. In the spatialGE web application, users can apply three clustering/domain detection methods, all of which explicitly incorporate the spatial information modality:

- **STclust ([Ospina et al. 2022](#)):**

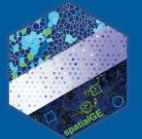
The built-in method in spatialGE, which groups spots/cells based on gene expression weighted by spatial Euclidean distances (*see page 2*).

- **SpaGCN ([Hu et al. 2021](#)):**

A domain detection method using graph convolutional networks (GCNs) to integrate gene expression and spatial coordinates, conducting clustering of the spots/cells (*see page 7*).

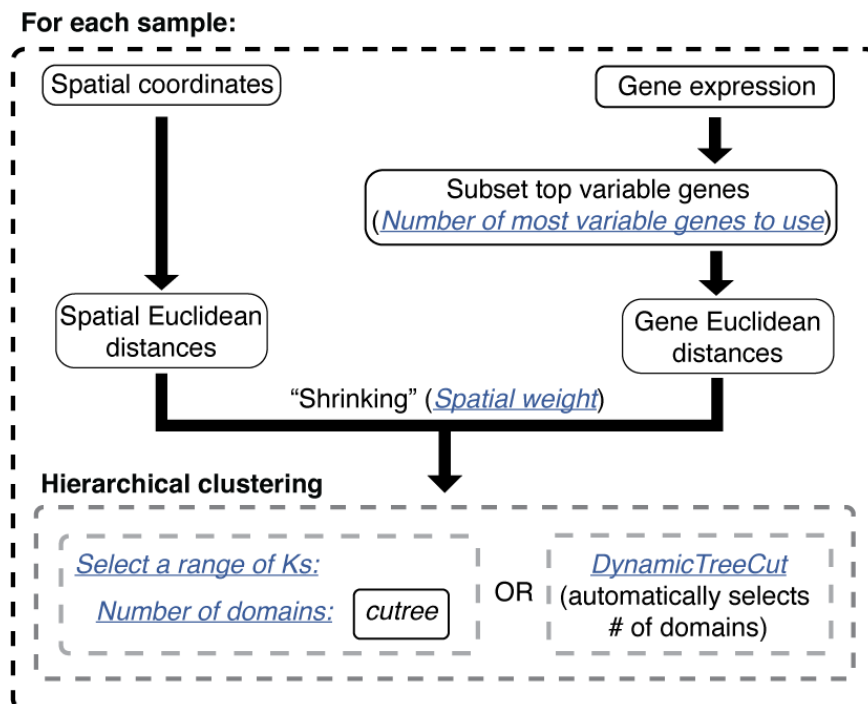
- **MILWRM ([Kaur et al. 2024](#)):**

A method to identify transcriptomically similar spatial domains across multiple samples. The employs batch correction and integration of gene expression and spatial information (*see page 13*).



STclust

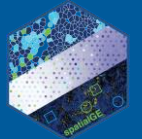
The STclust method is part of the **Spatial Domain Detection** module. It uses spatial distances among spots or cells to inform clustering, resulting in continuous clusters that can be considered tissue domains. This method applies hierarchical clustering to gene expression data, which has been adjusted ("shrunk") using spatial information in the form of Euclidean distances (details in [Ospina et al. 2022](#)). The diagram below explains the STclust algorithm:



For each sample, STclust calculates Euclidean distances among the spots/cells based on their spatial (x, y) locations. It also calculates Euclidean distances from gene expression using the top variable genes within that sample (**Number of most variable genes to use** slider). Next, the weighted average between the spatial distances and the gene expression distances is calculated. The weights are derived from the **Spatial weight** parameter, indicating the importance of spatial distances. These weighted distances are used in hierarchical clustering. Finally, the user can allow DynamicTreeCuts (Langfelder et al. 2008) to automatically decide the number of domains in the sample or define domains across a range of values (e.g., find two domains (k=2), three domains (k=3), four domains (k=4), etc.).

Some aspects to consider when running the **Spatial domain detection** module:

- STclust works on a sample-by-sample basis, and no cross-sample domains are identified. For example, cluster/domain "1" in sample_A may not be the same as cluster/domain "1" in sample_B. Differential expression analysis can be conducted to assign biological identities to clusters/domains and ascertain similarities across samples. Alternatively, the method MILWRM (*see page 13*) can be used for cross-sample domain identification.



- Selecting a value for the **Spatial weight** parameter is largely arbitrary. Users should try different combinations of spatial weight with a broad range of Ks (**Select a range of Ks**) or **DeepSplit** (if using **DynamicTreeCuts**). The decision on the best number of domains can also be facilitated by inspecting the corresponding tissue image. For reasonable results that resemble the overall tissue organization, spatial weights as low as 0.025 are often sufficient. Spatial weights over 0.1 might lead to non-biologically informative results, as domains will mostly reflect physical distances among spots/cells.
- The **DeepSplit** Parameter parameter provides rough control over the number of domains automatically detected with **DynamicTreeCuts**. Generally, the larger this value, the more domains are obtained.

To begin domain detection, start by defining a **Spatial weight**. For this example, the default value of 0.02 will be used. In addition to running STclust with Spatial weight=0.02, the Spatial domain detection module will always run with **Spatial weight**=0 as well. This allows the user to assess whether adding a spatial weight is necessary.

spatialGE

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Spatial Domain Detection with STclust

The algorithm STclust detects spatial domains via hierarchical clustering of gene expression weighted by spot-to-spot distances. Users have the option of specifying a number of clusters (k) or use automatic detection using DynamicTreeCuts. The number of clusters inferred by DynamicTreeCuts can be controlled via DeepSplit.

1 Spatial weight: 0.02

☒ Select a range of Ks ☐ Use DynamicTreeCuts

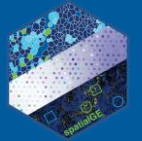
Number of domains: 2 5

Number of most variable genes to use: 3000

Next, select the method for detecting domains. There are two options available:

1. **Select a range of Ks**: This option allows you to define the number of expected domains by specifying a range of K values. To activate this option, click the **Select a range of Ks** radio button.
2. **Use DynamicTreeCuts**: This option uses the [R package DynamicTreeCut](#) to predict the number of domains automatically. To activate it, click the **Use DynamicTreeCuts** radio button.

For this tutorial, we will select the **Select a range of Ks** option, which enables the **Number of domains** slider. We will keep the range from 2 to 5 domains. This setting allows STclust to detect two, three, four, and five clusters/domains in each sample. It is up to the user to decide which number of clusters best explains the biology



of the samples. For a more informed decision, users are encouraged to perform [Differential expression](#) analysis, which is available as another module in spatialGE after **Spatial domain detection** is completed.

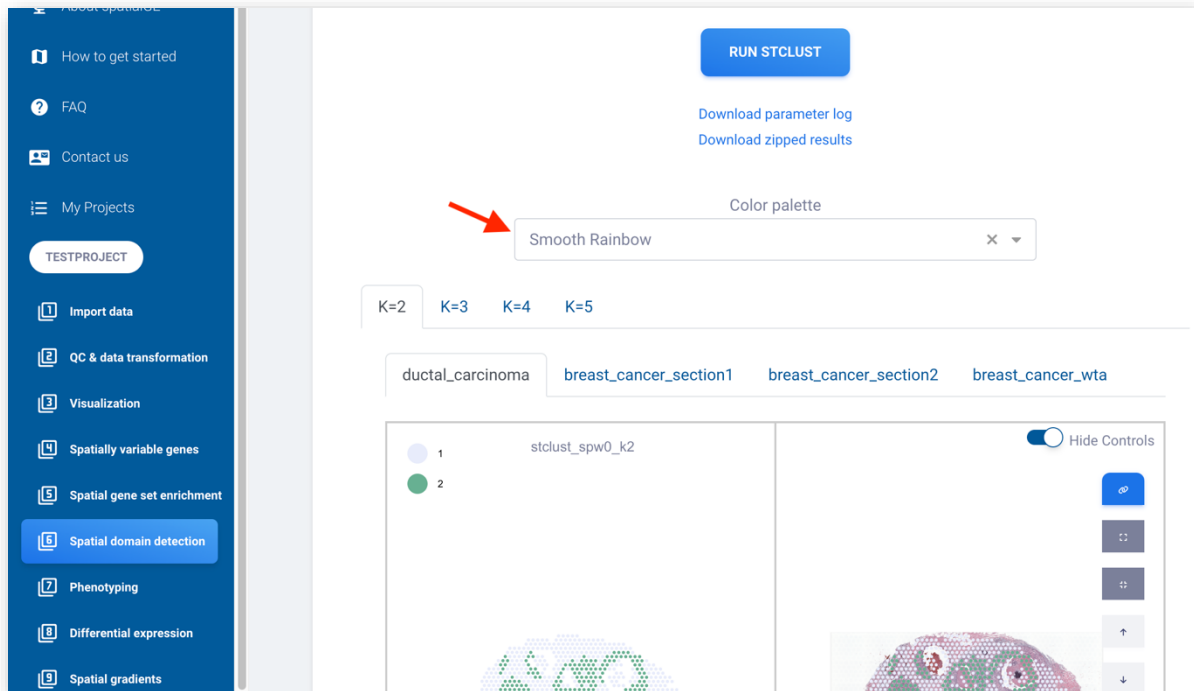
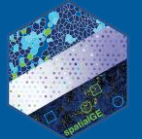
Note: If the user decides to use DynamicTreeCut, a single clustering solution will be presented per sample. After clicking the **Use DynamicTreeCuts** radio button, the **DeepSplit** slider will appear. The **DeepSplit** parameter controls the level of splitting that DynamicTreeCut performs. The higher the DeepSplit value, the more domains will be predicted per sample.

Now, specify the number of genes to input into STclust in the **Number of most variable genes to use** field. Genes are selected based on variance, estimated using the vst method from the R package Seurat. The default setting of 3000 genes should work for most cases. However, for this tutorial, we will set the number to 5000 genes to improve the accuracy of domain prediction. Note that more genes do not always lead to better predictions. Spatial transcriptomics data generally presents an excess of zeroes, and inputting too many genes can add noise, which may dampen the biological differences between tissue domains. Enter the number 5000 in the textbox above the **Number of most variable genes to use** slider. Finally, click the **RUN STCLUST** button to perform tissue domain detection.

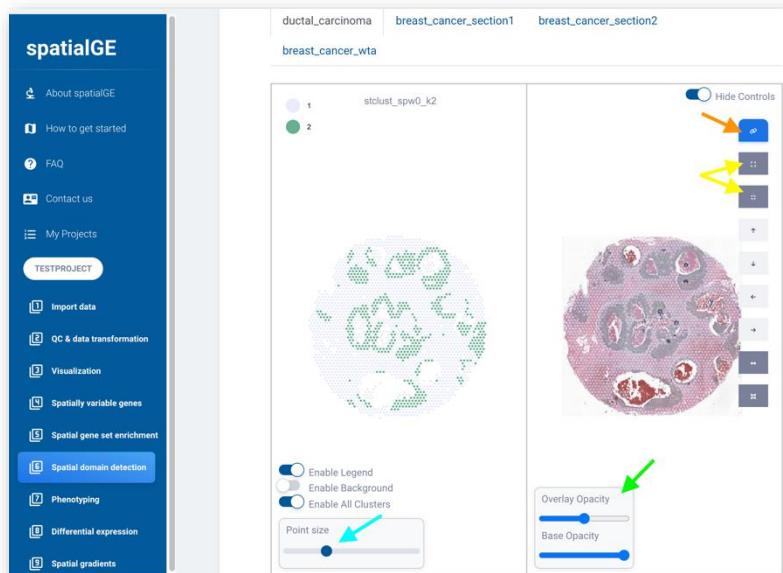
The screenshot shows the 'Spatial Domain Detection with STclust' module in the spatialGE interface. The sidebar on the left contains navigation links: 'About spatialGE', 'How to get started', 'FAQ', 'Contact us', 'My Projects', 'TESTPROJECT', 'Import data', 'QC & data transformation', 'Visualization', 'Spatially variable genes', 'Spatial gene set enrichment', 'Spatial domain detection' (highlighted), 'Phenotyping', and 'Differential expression'. The main panel displays the following configuration options:

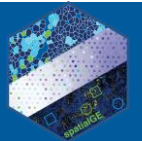
- Spatial weight:** 0.02 (with a help icon)
- Method:** ☒ Select a range of Ks (with a help icon) and ☐ Use DynamicTreeCuts (with a help icon)
- Number of domains:** A range from 2 to 5 (with a help icon)
- Number of most variable genes to use:** A text input field containing '5000' (with a help icon) and a slider below it. A red circle with the number '2' is next to the input field.
- RUN STCLUST:** A blue button with a red circle and the number '3' next to it.

Once STclust is finished, a series of tabs (K=2 to K=5) will appear below the **RUN STCLUST** button. Each tab presents the spatial locations of domains within the tissue samples (and the tissue images, if available). Alternative color palettes are available, generated using the [Khroma](#) and [RColorBrewer](#) R packages (red arrow). Completing analysis with the **Spatial domain detection** module enables the [Differential expression](#) and [Spatial gradients](#) modules.



Within each of the “K” tabs, there are additional sub-tabs containing the results for each sample. These tabs show the results from STclust using a spatial weight=0. Scrolling down to observe the results of spatial weight=0.02. The resulting spatial plots can be modified using several controls in the interface. Some of the most important controls include “Point size” to modify the size of the spots or cells (turquoise arrow), opacity of the plot and tissue image (if available, green arrow), and adjustment of plot to tissue image (manual image registration, yellow arrow). Finally, images can be exported as PDF/PNG/SVG first by clicking the “Synchronize” button (orange arrow) and then the “Export” button (three-bar icon appearing after clicking the “Synchronize” button).





Additionally, results from differential expression analysis among domains are shown in table format. Importantly, the differential expression analysis within the **Spatial domain detection** module is conducted only for the top 50 most variable genes. Users must use the [Differential expression](#) module for a complete analysis of the identified domains.

The **Spatial domain detection** module also allows manual renaming of the identified domains. From the differential expression results in this example, it can be observed that spots assigned to spatial domain “1” express two immunoglobulins (*IGHA1* and *IGHG2*). The original name “1” can be changed by typing in the corresponding text box (here changed to “immune_infiltrated”, blue arrow). The name of the new manual annotation can be changed in the “Modified annotation” text box (here changed to “user_annotation”, pink arrow). Make sure to save the changes of the manual annotation (“SAVE CHANGES” orange button).

Original annotation: stclust_spw0_k2
Modified annotation: user_annotation

Clusters:

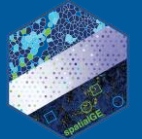
Original	Modified
1	immune_infiltrated
2	2

Changes detected, please be sure to save them when ready

SAVE CHANGES

Gene	Average log-Fold Change	Cluster 1
IGHG2	1.748	1
IGHA1	1.572	1
JCHAIN	1.387	1
HILPDA	-0.993	1

Upon saving, manually re-annotated spatial domains are available for use in the [Differential expression](#) and [Spatial gradients](#) modules.



SpaGCN

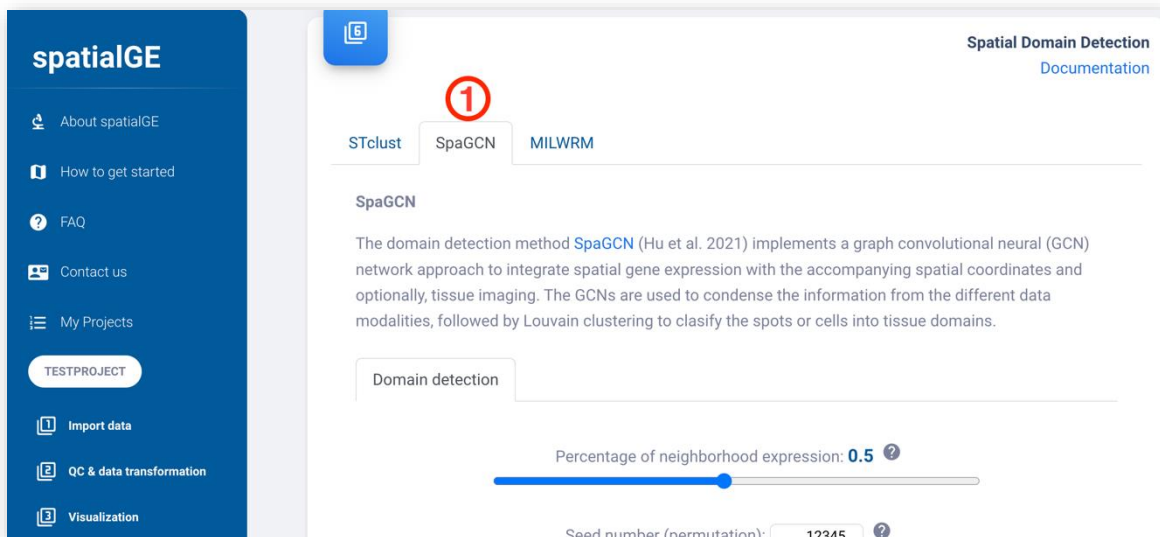
The SpaGCN method ([Hu et al. 2021](#)) is implemented in the **Spatial domain detection** module. SpaGCN uses graph convolutional networks (GCNs) to perform unsupervised clustering of spots or cells in a spatial transcriptomics (ST) sample. In brief, SpaGCN constructs a graph from spatial coordinates to describe the spatial relationships between spots/cells, and a GCN is constructed to summarize gene expression of neighboring spots/cells. Domains are identified from the GCN using Louvain clustering followed by iterative clustering. For datasets containing Visium samples, users can optionally refine domain assignments to achieve more spatially continuous domains (see [Hu et al. 2021](#) for details). Following domain detection, SpaGCN also allows for spatially variable gene (SVG) testing.

Some additional aspects to consider about the spatialGE implementation of SpaGCN:

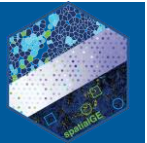
- Tissue images: The command line implementation of SpaGCN supports using tissue images to better inform spatial relationships. This feature is not yet implemented in the **Spatial domain detection** module of spatialGE but will be available in the future.
- Seed number: Users are encouraged to run SpaGCN with different seed values and save the results to check for consistency in tissue domain predictions.

SpaGCN – Domain Detection

To begin domain detection, click the **SpaGCN** tab next to the STclust tab.

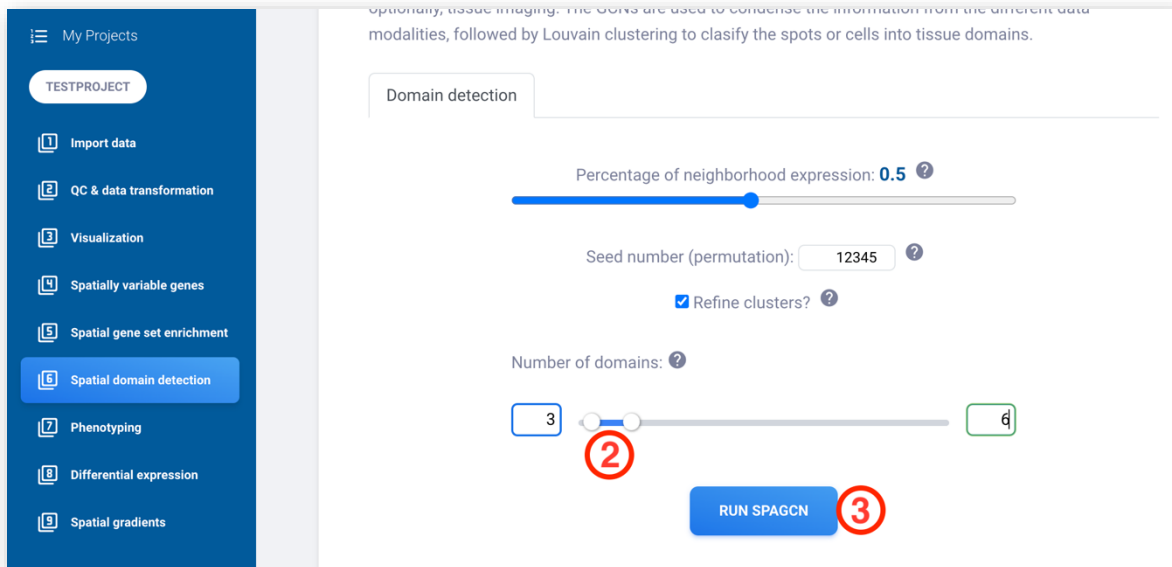


When running SpaGCN, one important parameter to define is the **Percentage of neighborhood expression** contributing to the expression profile of a given spot or cell. The authors of SpaGCN recommend setting this parameter to 0.5 (default) for Visium experiments. However, if using data from single-cell spatial transcriptomics, this parameter should be set higher to ensure the summarized expression is more influenced by neighboring cells.

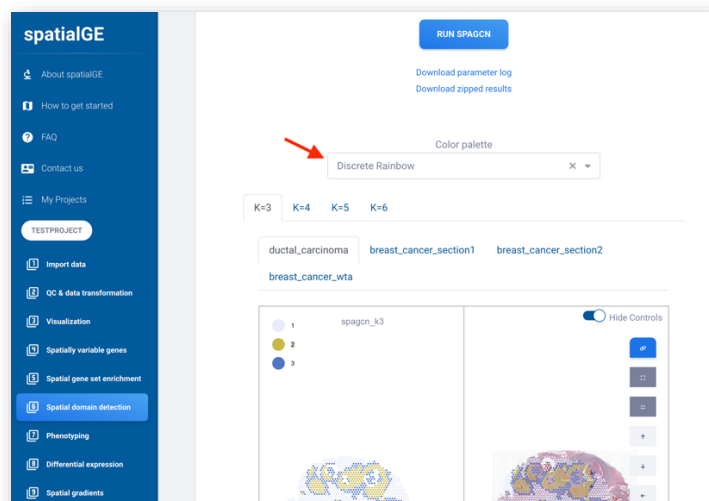


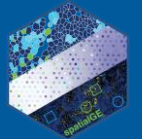
Since the dataset used in this example was generated with Visium (see [Import data](#)), we will leave **Percentage of neighborhood expression** parameter at its default value of 0.5.

Next, use the **Number of domains** slider to specify a range of K values (i.e., the number of expected domains). SpaGCN will be executed for each selected K value. For this tutorial, set the slider so that the range of K values is from 3 to 6. Finally, click the **RUN SPAGCN** button to perform tissue domain detection.

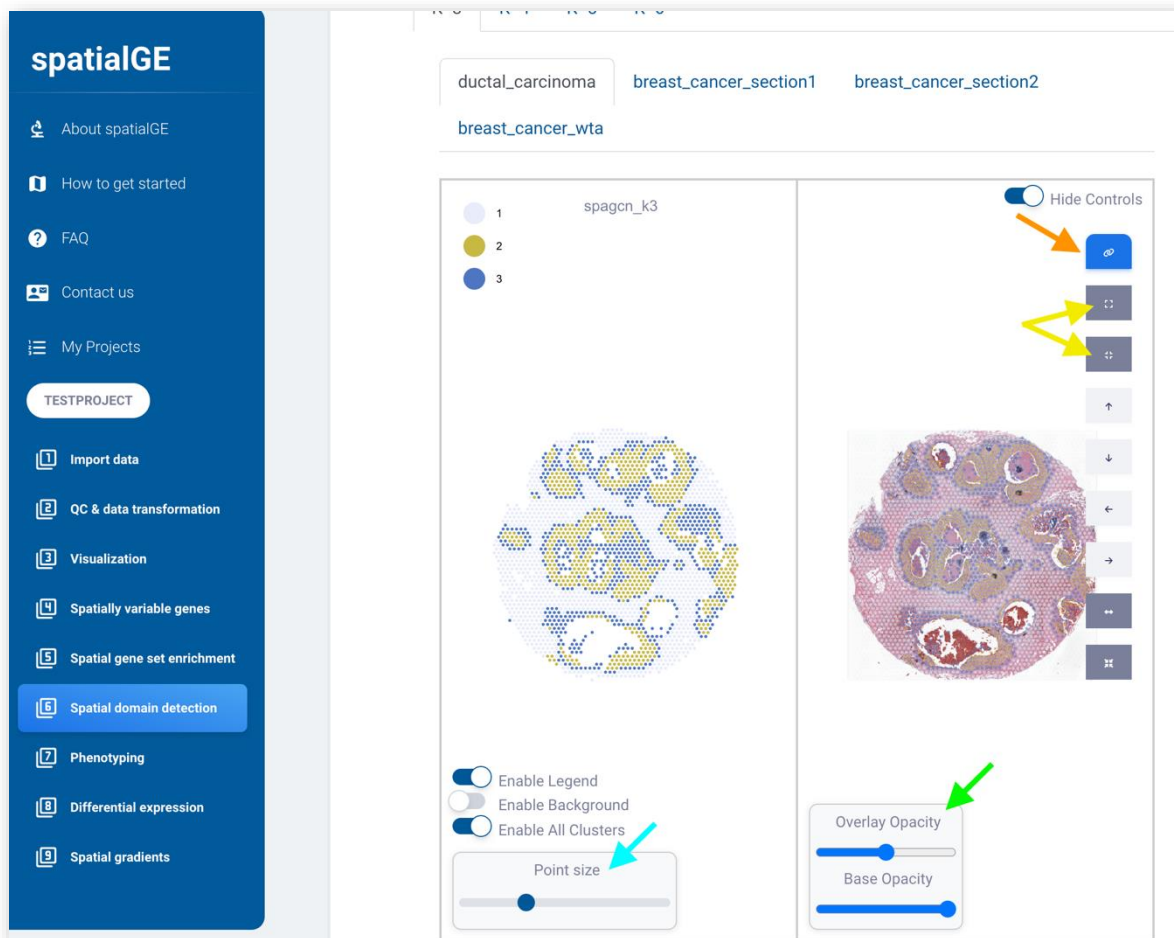


Once STclust is finished, a series of tabs (K=3 to K=5) will appear below the **RUN SPAGCN** button. Each tab presents the spatial locations of domains within the tissue samples (and the tissue images, if available). Alternative color palettes are available, generated using the [Khroma](#) and [RColorBrewer](#) R packages (red arrow). Completing analysis with the **Spatial domain detection** module enables the [Differential expression](#) and [Spatial gradients](#) modules.





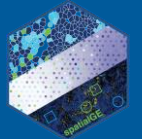
Within each of the “K” tabs, there are additional sub-tabs containing the results for each sample. If the **Refine clusters** checkbox was activated, scroll down to observe the results of *cluster refinement* (see [Hu et al. 2021](#) for details). The resulting spatial plots can be modified using several controls in the interface. Some of the most important controls include “Point size” to modify the size of the spots or cells (turquoise arrow), opacity of the plot and tissue image (if available, green arrow), and adjustment of plot to tissue image (manual image registration, yellow arrow). Finally, images can be exported as PDF/PNG/SVG first by clicking the “Synchronize” button (orange arrow) and then the “Export” button (three-bar icon appearing after clicking the “Synchronize” button).



Additionally, results from differential expression analysis among domains are shown in table format, however:

- The differential expression analysis within the **Spatial domain detection** module is conducted only for the top 50 most variable genes. Users must use the **Differential expression** module for a complete analysis of the identified domains.
- The results are generated via Wilcoxon’s tests. These results differ from the SVG tests that SpaGCN can generate (see below: “SpaGCN – Spatially variable genes”).

The **Spatial domain detection** module also allows manual renaming of the identified domains. From the differential expression results in this example, it can be observed that spots assigned to spatial domain “1” express



two immunoglobulins (*IGHA1* and *IGHG2*). The original name “1” can be changed by typing in the corresponding text box (here changed to “immune_infiltrated”, blue arrow). The name of the new manual annotation can be changed in the “Modified annotation” text box (here changed to “spagcn_k3_manual”, pink arrow). Make sure to save the changes of the manual annotation (“SAVE CHANGES” orange button).

The screenshot shows the spatialGE web interface. On the left is a sidebar with navigation links: About spatialGE, How to get started, FAQ, Contact us, My Projects, TESTPROJECT, Import data, QC & data transformation, Visualization, Spatially variable genes, Spatial gene set enrichment, **Spatial domain detection**, Phenotyping, Differential expression, and Spatial gradients. The main content area displays the 'Spatial domain detection' module. It shows the 'Original annotation: spagcn_k3' and the 'Modified annotation: spagcn_k3_manual' (highlighted with a pink arrow). Below this is a table with columns 'Original' and 'Modified'. The first row shows '1' in the 'Original' column and 'immune_infiltrated' in the 'Modified' column (highlighted with a blue arrow). The second row shows '3' in both columns. The third row shows '2' in both columns. Below the table is a message: 'Changes detected, please be sure to save them when ready' and a 'SAVE CHANGES' button. At the bottom is a table with columns 'Gene', 'Average log-Fold Change', and 'Clus'. The table contains the following data:

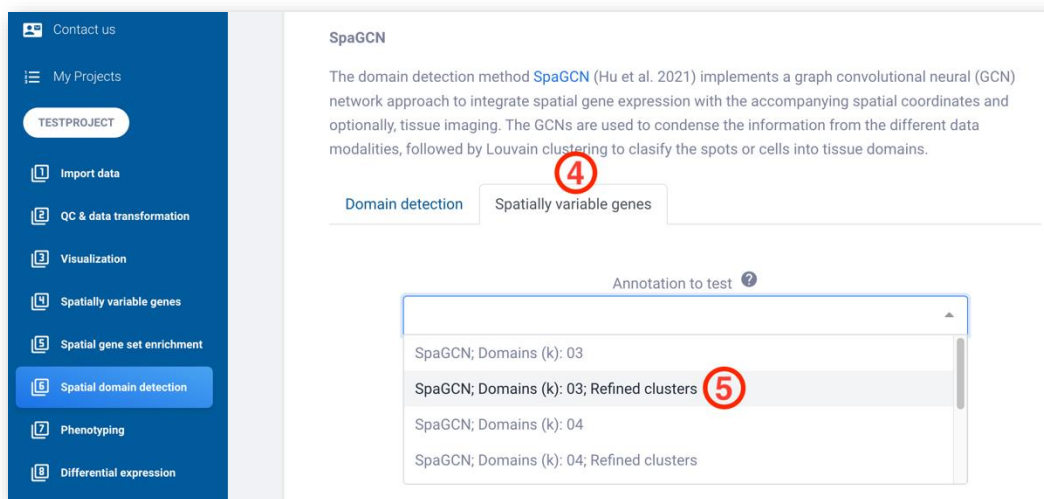
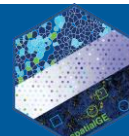
Gene	Average log-Fold Change	Clus
JCHAIN	1.438	
IGHG2	1.502	
LTF	-1.091	
IGHA1	1.399	
LCN2	-0.978	

Upon saving, manually re-annotated spatial domains are available for use in the [Differential expression](#) and [Spatial gradients](#) modules.

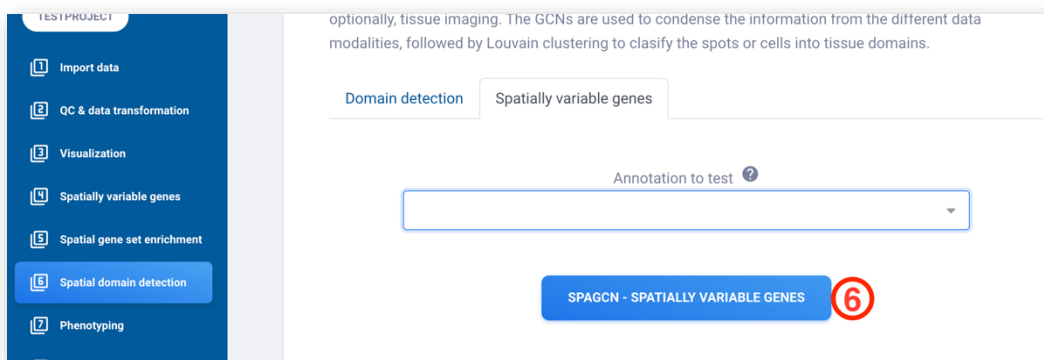
SpaGCN – Spatially Variable Genes

With spatial domains inferred using SpaGCN, the **Spatially variable genes** tab is now enabled. To start detecting SVGs, click on this tab. This section includes a single drop-down menu labeled “**Annotation to test**” which allows the user to select the domain detection solution that best reflects the tissue's biology. In practice, the user should run the SVG detection multiple times, selecting different domain detection solutions and saving the results each time.

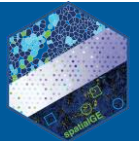
In this example, SVGs will be detected using the “*SpaGCN; Domains (k): 03; Refined clusters*” option in the **Annotation to test** dropdown. By selecting this option, SpaGCN will identify SVGs for each of the previously inferred and refined clusters (Domains 1, 2, and 3).



Next, click the **SPAGCN – SPATIALLY VARIABLE GENES** button to start the analysis.



After SVG detection is completed, a series of tabs will appear below the **SPAGCN – SPATIALLY VARIABLE GENES** button. The first level of tabs contains SVGs for each sample. Within each sample tab, SVGs for each domain are displayed (three tabs in this case, since $K=3$ was selected in the **Annotation to test** dropdown). For convenience, the gene symbols are links to the GeneCards database, providing information about each gene.



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3 Visualization

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5 Spatial gene set enrichment

Spatial domain detection

7 Phenotyping

8 Differential expression

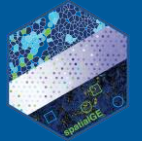
9 Spatial gradients

ductal_carcinoma breast_cancer_section1 breast_cancer_section2 breast_cancer_wta

Domain 1 Domain 2 Domain 3

CONTINUOUS SCROLLING

☐	Gene	In-grou...	Out-gro...	In/out-g...	In-grou...	Out-gro...	Fold ch...	Adjuste...
	Q	Q	Q	Q	Q	Q	Q	Q
☐	PPDPF	1	1	1	4.049	3.314	2.084	8.14e-181
☐	VIM	0.988	0.994	0.993	1.626	2.93	0.271	1.253e-...
☐	COL1A1	1	1	1	2.971	4.41	0.237	7.607e-...
☐	COL1A2	0.998	1	0.998	2.667	4.08	0.243	4.934e-...
☐	COL3A1	0.998	0.997	1.001	2.37	3.768	0.247	2.234e-...
☐	CST3	0.912	0.977	0.934	1.051	2.15	0.333	2.041e-...
☐	MAL2	1	0.957	1.045	2.768	1.868	2.459	7.125e-...
☐	ERBB2	1	0.997	1.003	3.525	2.683	2.32	7.999e-...
☐	KRT7	0.998	0.987	1.012	3.179	2.297	2.415	1.41e-151
☐	FTL	1	1	1	3.695	4.859	0.312	4.748e-...

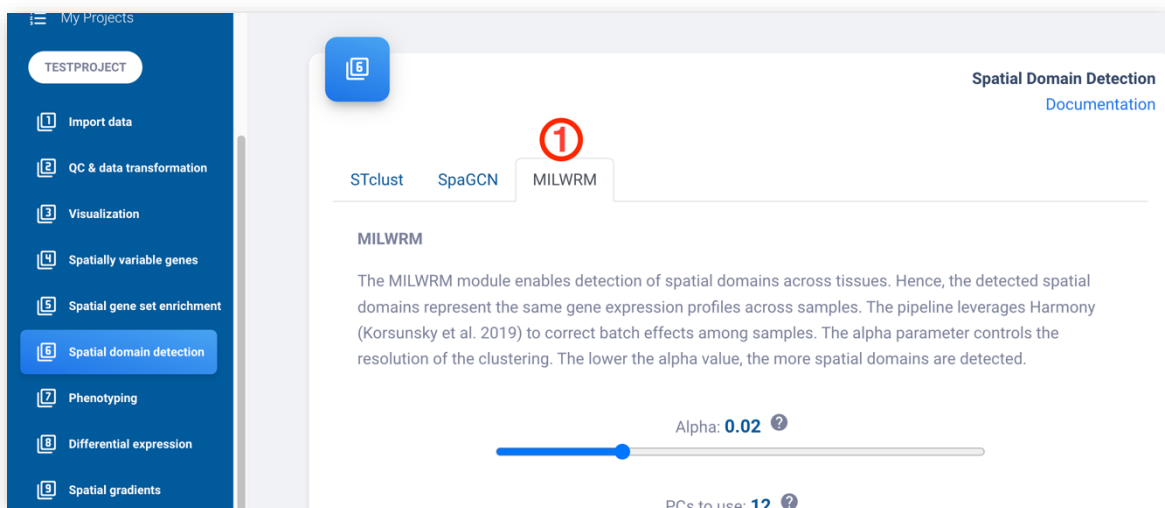


MILWRM

The MILWRM method ([Kaur et al. 2024](#)) is implemented in the **Spatial domain detection** module. Although MILWRM has been designed for use with different spatial omics, the spatial transcriptomics (ST) workflow is the implementation used in spatialGE. The implementation leverages batch effect correction via Harmony ([Korsunsky et al. 2019](#)) to facilitate the identification of spatial domains comparable across ST samples.

Currently, **MILWRM is enabled in spatialGE exclusively for Visium data sets**. Future implementations might include a MILWRM workflow for single-cell ST data sets.

To begin domain detection, click the **MILWRM** tab next to the SpaGCN tab.

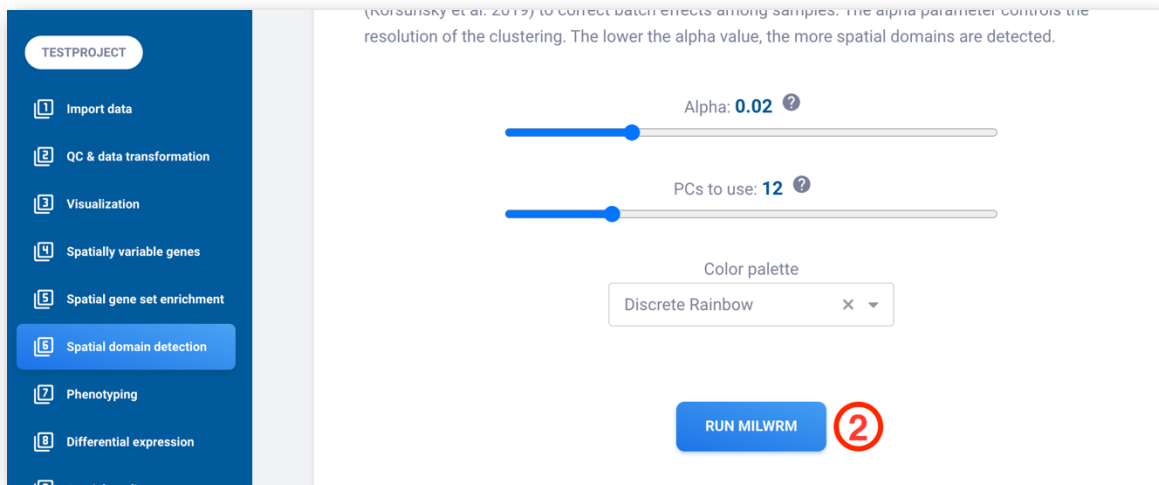
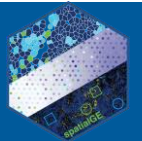


At the core of MILWRM, tissue domains are identified using K-means clustering. The number of domains is selected based on adjusted inertia ([Kaur et al. 2024](#)), which penalizes clustering solutions with a high number of domains. Users can control this penalization with the **Alpha** parameter slider; higher alpha values favor solutions with fewer domains.

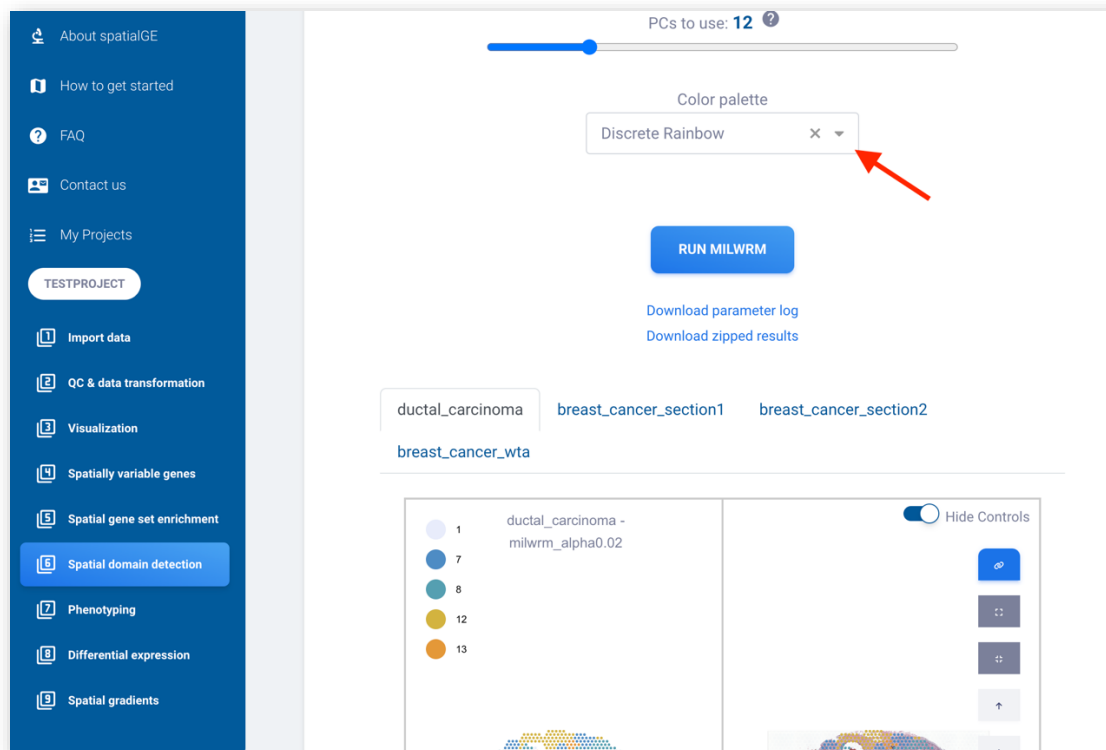
In spatialGE, MILWRM allows for alpha values ranging from 0.01 to 0.05 (default is 0.02). Additionally, this implementation uses dimensionality reduction (PCA) instead of normalized gene expression as input for K-means. Users can adjust the number of principal components (PCs) used as input with the **PCs to use** slider.

For this example, the two values will be kept at their defaults: **Alpha**=0.02 and **PCs to use**=12. Nonetheless, users are encouraged to save the results and re-run MILWRM with different values, especially the number of PCs, which indirectly controls the amount of gene expression information used during clustering.

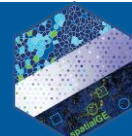
To initiate MILWRM, click the **RUN MILWRM** button.



Once MILWRM is finished, a series of tabs will appear below the **RUN MILWRM** button. Each tab presents the spatial locations of domains within the tissue samples (and the tissue images, if available). Alternative color palettes are available, generated using the [Khroma](#) and [RColorBrewer](#) R packages (red arrow). Completing analysis with the **Spatial domain detection** module enables the **Differential expression** and **Spatial gradients** modules.



The resulting spatial plots can be modified using several controls in the interface. Some of the most important controls include “Point size” to modify the size of the spots or cells (turquoise arrow), opacity of the plot and



tissue image (if available, green arrow), and adjustment of plot to tissue image (manual image registration, yellow arrow). Finally, images can be exported as PDF/PNG/SVG first by clicking the “Synchronize” button (orange arrow) and then the “Export” button (three-bar icon appearing after clicking the “Synchronize” button).



Finally, clicking a cluster color in the legend hides the corresponding spots in the plot (for example, Cluster 7, pink arrow):

